Inequalities

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1 Hölder inequality

A classic result concerning p-norms is the **Hölder inequality in inner-product form**:

$$|x^T y| \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Proof. Recall the definition of dual norm,

$$\|x\|_p = \max_{\|z\|_q \leq 1} x^T z.$$

Thus, $x^T z \leq \|x\|_p$ holds for any z satisfying $\|z\|_q \leq 1$, including $\|z\|_q = 1$. When $\|z\|_q = 1$, we have

$$x^T z \leq \|x\|_p \|z\|_q$$

Now let $z = ty$ with $t > 0$. Thus, we have

$$x^T (ty) \leq \|x\|_p \|ty\|_q \Longleftrightarrow x^T y \leq \|x\|_p \|y\|_q$$

where $\|y\|_q = \|z/t\|_q = 1/t \|z\|_q = 1/t > 0$. This completes the proof. □

The key part of the above proof is using the dual representation of the ℓ_p norm, namely, $\|x\|_p = \max_{\|z\|_q \leq 1} x^T z$.

1.1 Cauchy-Schwarz inequality

A very important special case of Hölder inequality is **Cauchy-Schwarz inequality**:

$$|x^T y| \leq \|x\|_2 \|y\|_2.$$

which can also be expressed as $(x^T y)^2 \leq \|x\|_2^2 \|y\|_2^2$. Here, $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$. Thus,

$$(x_1 y_1 + x_2 y_2 + \ldots + x_n y_n)^2 \leq (x_1^2 + x_2^2 + \ldots + x_n^2)(y_1^2 + y_2^2 + \ldots + y_n^2)$$
$$(\sum_{i=1}^n x_i y_i)^2 = \sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i^2$$

1.2 Two variants of Cauchy-Schwarz inequality

Given $a_i \in \mathbf{R}$ and $b_i > 0 (i = 1, 2, \cdots, n)$, let $y_i = \sqrt{b_i}$ and $x_i = \frac{a_i}{\sqrt{b_i}}$. Then we have an important **variant** of Cauchy-Schwarz inequality.

$$(\sum_{i=1}^n \frac{a_i}{\sqrt{b_i}} \sqrt{b_i})^2 = (\sum_{i=1}^n a_i)^2 \leq (\sum_{i=1}^n \frac{a_i^2}{b_i}) \sum_{i=1}^n b_i \Longleftrightarrow \sum_{i=1}^n \frac{a_i^2}{b_i} \geq \frac{(\sum_{i=1}^n a_i)^2}{\sum_{i=1}^n b_i}.$$

For $a_i > 0$ and $b_i > 0 (i = 1, 2, \cdots, n)$, letting $x_i = \sqrt{\frac{a_i}{b_i}}$ and $y_i = \sqrt{a_i b_i}$ gives

$$\sum_{i=1}^n x_i^2 \cdot \sum_{i=1}^n y_i^2 \geq (\sum_{i=1}^n x_i y_i)^2 \Longleftrightarrow \sum_{i=1}^n \frac{a_i}{b_i} \cdot \sum_{i=1}^n a_i b_i \geq (\sum_{i=1}^n a_i)^2 \Longleftrightarrow \sum_{i=1}^n \frac{a_i}{b_i} \geq \frac{(\sum_{i=1}^n a_i)^2}{\sum_{i=1}^n a_i b_i}.$$

2 Rearrangement inequalities

For two sequences $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$ with reals a_i and b_i, the following inequalities

$$\underbrace{a_1 b_n + a_2 b_{n-1} + \cdots + a_n b_1}_{\text{reverse sum}} \leq \underbrace{a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}}_{\text{disordered sum}} \leq \underbrace{a_1 b_1 + a_2 b_2 + \cdots + a_n b_n}_{\text{sequential sum}}$$

hold, where $\pi(1), \pi(2), \ldots, \pi(n)$ is any permutation of $1, 2, \ldots, n$. Simply speaking, **reverse sum $\leq$ disordered sum $\leq$ sequential sum**.

Proof. [1] We want to prove that the identity permutation maximizes $a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$. Suppose for the sake of contradiction $\pi(i)$ is the smallest integer such that $\pi(i) \neq i$, then $\pi(i) = j (j > i)$ (since $1, \ldots, i-1$ have been assigned). Meanwhile, there exists a number $k > i$ such that $\pi(k) = i$ as some number must be assigned to i.

Now, since $i < j$, it follows that $b_i \leq b_j$. Likewise, since $i < k$, it follows that $a_i \leq a_k$. Thus,

$$(a_k - a_i)(b_j - b_i) \geq 0 \Longrightarrow a_k b_j + a_i b_i \geq a_i b_j + a_k b_i,$$

which demonstrates that the sum $a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$ is not decreased by changing $\pi(j) = i$ and $\pi(k) = i$ to $\pi(i) = i$ and $\pi(k) = j$. This implies the identity permutation gives the maximum possible value of the sum $a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$, as desired. Moreover, the reverse identity permutation gives the minimum possible value of the sum $a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$. $\square$

[1] https://brilliant.org/wiki/rearrangement-inequality/

2.1 Chebyshev's inequality

For two sequences $a_1 \le a_2 \le \cdots \le a_n$ and $b_1 \le b_2 \le \cdots \le b_n$ with reals a_i and b_i, by the rearrangement inequalities, we always have the following inequalities

$$\begin{gathered}
x_1y_n + x_2y_{n-1} + \cdots + x_ny_1 \le x_1y_1 + x_2y_2 + \cdots + x_ny_n \le x_1y_1 + x_2y_2 + \cdots + x_ny_n \\
x_1y_n + x_2y_{n-1} + \cdots + x_ny_1 \le x_1y_2 + x_2y_3 + \cdots + x_ny_1 \le x_1y_1 + x_2y_2 + \cdots + x_ny_n \\
x_1y_n + x_2y_{n-1} + \cdots + x_ny_1 \le x_1y_3 + x_2y_4 + \cdots + x_ny_2 \le x_1y_1 + x_2y_2 + \cdots + x_ny_n \\
\cdots \\
x_1y_n + x_2y_{n-1} + \cdots + x_ny_1 \le x_1y_n + x_2y_1 + \cdots + x_ny_{n-1} \le x_1y_1 + x_2y_2 + \cdots + x_ny_n
\end{gathered}$$

Summing up the above inequalities gives

$$\begin{gathered}
n(x_1y_n + x_2y_{n-1} + \cdots + x_ny_1) \le (x_1 + x_2 + \cdots + x_n)(y_1 + y_2 + \cdots + y_n) \le n(x_1y_1 + x_2y_2 + \cdots + x_ny_n) \\
\frac{1}{n}(x_1y_n + x_2y_{n-1} + \cdots + x_ny_1) \le \frac{1}{n^2}(x_1 + x_2 + \cdots + x_n)(y_1 + y_2 + \cdots + y_n) \le \frac{1}{n}(x_1y_1 + x_2y_2 + \cdots + x_ny_n) \\
\frac{1}{n}\sum_{i=1}^{n} x_iy_{n+1-i} \le \frac{1}{n}\sum_{i=1}^{n} x_i \cdot \frac{1}{n}\sum_{i=1}^{n} y_i \le \frac{1}{n}\sum_{i=1}^{n} x_iy_i
\end{gathered}$$

which is called Chebyshev's inequality.

3 Quadratic mean, average mean, geometric mean, and harmonic mean

The content in this section is largely taken from Xicheng Peng et al. Exploring Inequalities. 2016. In this section, we give the relationships between quadratic mean (QM), average mean (AM), geometric mean (GM), and harmonic mean (HM). The result is QM≥AM≥GM≥HM.

For any positive real number $a_1, a_2, \cdots, a_n$, we have the following inequalities

$$\overbrace{\sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}}}^{\text{Quadratic Mean}} \ge \overbrace{\frac{a_1 + a_2 + \cdots + a_n}{n}}^{\text{Arithmetic Mean}} \ge \overbrace{\sqrt[n]{a_1a_2\cdots a_n}}^{\text{Geometric Mean}} \ge \overbrace{\frac{n}{\frac{1}{a_1} + \frac{1}{a_1} + \cdots + \frac{1}{a_n}}}^{\text{Harmonic Mean}} \tag{1}$$

where the equalities hold if and only if $a_1 = a_2 = \cdots = a_n$. This inequality chain is also denoted as $H(n) \le G(n) \le A(n) \le Q(n)$.

We will provide two proofs. The first proof employs Jensen's inequality and is simpler compared with the second proof.

Proof. Let $f(x) = -\ln x$, then $f''(x) = \frac{1}{x^2} > 0$ which implies that f is convex. For all $a_i > 0$, by Jensen's inequality,

$$\begin{gathered}
f\left(\frac{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}{n}\right) \le \frac{1}{n}\left(f(\frac{1}{a_1}) + f(\frac{1}{a_2}) + \cdots + f(\frac{1}{a_n})\right) \\
-\ln\left(\frac{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}{n}\right) \le \frac{1}{n}\left(-\ln(\frac{1}{a_1}) - \ln(\frac{1}{a_2}) + \cdots - \ln(\frac{1}{a_n})\right) \\
\ln\left(\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}\right) \le \frac{1}{n}(\ln a_1 + \ln a_2 + \cdots + \ln a_n) \\
\ln\left(\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}\right) \le \ln \sqrt[n]{a_1a_2\cdots a_n}
\end{gathered}$$

Since ln is increasing,

$$\frac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\cdots+\frac{1}{a_n}} \leq \sqrt[n]{a_1a_2\cdots a_n}$$

where the equality holds if and only if $a_1 = a_2 = \cdots = a_n$. The "if" part is obvious. For the "only if" part, suppose the equality holds with $a_1 \neq a_2 = a_3 = \cdots = a_n$, then let $a_1 = ka_2 (k > 0)$ we have

$$\frac{n}{(n-1+\frac{1}{k})\frac{1}{a_2}} = a_2\sqrt[n]{k} \Longleftrightarrow \frac{n}{(n-1+\frac{1}{k})} = \sqrt[n]{k} \Longleftrightarrow n = \sqrt[n]{k}(n-1+\frac{1}{k})$$

Let $x = \sqrt[n]{k}$ and it is clear that $x > 0$. Then

$$nx^{n-1} = (n-1)x^n + 1 \Longleftrightarrow (n-1)x^n - nx^{n-1} + 1 = 0$$

$$\begin{aligned}
(n-1)x^n - nx^{n-1} + 1 &= (n-1)x^n - (n-1)x^{n-1} - x^{n-1} + 1 \\
&= (n-1)x^{n-1}(x-1) - (x^{n-1}-1) \\
&= (n-1)x^{n-1}(x-1) - (x-1)(x^{n-2}+x^{n-3}+\cdots+1) \\
&= (x-1)[(n-1)x^{n-1} - (x^{n-2}+x^{n-3}+\cdots+1)] \\
&= (x-1)[(n-1)x^{n-1} - (n-1)x^{n-2} + (n-2)x^{n-2} - x^{n-3} - \cdots - 1] \\
&= (x-1)[(n-1)x^{n-2}(x-1) + (n-2)x^{n-2} - x^{n-3} - \cdots - 1] \\
&= (x-1)[(n-1)x^{n-2}(x-1) + (n-2)x^{n-2} - (n-2)x^{n-3} + (n-3)x^{n-3} - \cdots - 1] \\
&= (x-1)[(n-1)x^{n-2}(x-1) + (n-2)x^{n-3}(x-1) + (n-3)x^{n-3} - \cdots - 1] \\
&= (x-1)[(n-1)x^{n-2}(x-1) + (n-2)x^{n-3}(x-1) + (n-3)x^{n-4}(x-1) + \cdots + (x-1)] \\
&= (x-1)^2[(n-1)x^{n-2} + (n-2)x^{n-3} + (n-3)x^{n-4} + \cdots + 1]
\end{aligned}$$

Due to the fact that any polynomial that has positive coefficients cannot have roots on the nonnegative real axis[2], 1 is the only roots which contradicts the supposition that $a_1 \neq a_2$. Hence, $a_1 = a_2 = \cdots = a_n$ is the necessary and sufficient condition for the equality to hold.

Now we show GM≤AM,

$$\begin{aligned}
f(\frac{a_1+a_2+\cdots+a_n}{n}) &\leq \frac{1}{n}\left(f(a_1)+f(a_2)+\cdots+f(a_n)\right) \\
-\ln(\frac{a_1+a_2+\cdots+a_n}{n}) &\leq \frac{1}{n}\left(-\ln(a_1)-\ln(a_1)+\cdots-\ln(a_1)\right) \\
\ln(\frac{a_1+a_2+\cdots+a_n}{n}) &\geq \frac{1}{n}(\ln a_1 + \ln a_2 + \cdots + \ln a_n) \\
\ln(\frac{a_1+a_2+\cdots+a_n}{n}) &\geq \ln\sqrt[n]{a_1a_2\cdots a_n} \\
\frac{a_1+a_2+\cdots+a_n}{n} &\geq \sqrt[n]{a_1a_2\cdots a_n}
\end{aligned}$$

Let $g(x) = x^2$, then $g''(x) = 2 >$ which indicates that g is convex.

Now we show AM≤QM. By Jensen's inequality, we have

$$\begin{aligned}
g(\frac{a_1+a_2+\cdots+a_n}{n}) &\leq \frac{1}{n}\left(g(a_1)+g(a_2)+\cdots+g(a_n)\right) \\
(\frac{a_1+a_2+\cdots+a_n}{n})^2 &\leq \frac{1}{n}(a_1^2+a_2^2+\cdots+a_n^2)
\end{aligned}$$

[2] https://mtns2018.hkust.edu.hk/media/files/0073.pdf. Besides that paper, we can also prove this fact via contradiction. Suppose this kind of polynomial $P(x)$ has some positive real roots, say $P(x_0) = 0$, then each term of $P(x_0)$ is positive which leads to $P(x_0) > 0$, a contradiction. Thus, x_0 does not exist.

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \leq \sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}}$$

This completes our proof. $\square$

We first introduce a lemma for the final proof.

Lemma 3.1. *If $a_i > 0, i = 1, 2, \cdots, n$, and $a_1 a_2 \cdots a_n = 1$, then $a_1 + a_2 + \cdots + a_n \geq n$ where the equality holds iff $a_1 = a_2 = \cdots = a_n = 1$.*

Proof. Let $x_i = \ln a_i, i = 1, 2, \cdots, n$.

$$a_1 a_2 \cdots a_n = 1 \Longleftrightarrow \ln a_1 + \ln a_2 + \cdots + \ln a_n = 0 \Longleftrightarrow x_1 + x_2 + \cdots + x_n = 0$$

Since $e^x \geq x + 1$,

$$a_1 + a_2 + \cdots + a_n = e^{x_1} + e^{x_2} + \cdots + e^{x_n} \geq (x_1 + 1) + (x_2 + 1) + \cdots + (x_n + 1) = n$$

where the equality holds iff $x_i = 0, \forall i = 1, 2, \cdots, n$, i.e., $a_i = 1, \forall i = 1, 2, \cdots, n$ due to the fact that $e^x = x + 1$ iff $x = 0$. Thus, $a_1 + a_2 + \cdots + a_n = n$ iff $a_1 = a_2 = \cdots = a_n = 1$. $\square$

Next, we prove this inequality chain using Lemma 3.1.

Proof.

1. $H(n) \leq G(n)$

$$\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} \leq \sqrt[n]{a_1 a_2 \cdots a_n} \Longleftrightarrow \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_1} + \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_2} + \cdots + \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_n} \geq n$$

   We observe that $\frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_1} \cdot \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_2} \cdot \cdots \cdot \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_n} = 1$. By Lemma 3.1, the above holds. The equality holds iff $\frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_1} = \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_2} = \cdots = \frac{\sqrt[n]{a_1 a_2 \cdots a_n}}{a_n}$, i.e., $a_1 = a_2 = \cdots = a_n$.

2. $G(n) \leq A(n)$

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n} \Longleftrightarrow \frac{a_1}{\sqrt[n]{a_1 a_2 \cdots a_n}} + \frac{a_2}{\sqrt[n]{a_1 a_2 \cdots a_n}} + \cdots + \frac{a_n}{\sqrt[n]{a_1 a_2 \cdots a_n}} \geq n$$

   We observe that $\frac{a_1}{\sqrt[n]{a_1 a_2 \cdots a_n}} \cdot \frac{a_2}{\sqrt[n]{a_1 a_2 \cdots a_n}} \cdot \cdots \cdot \frac{a_n}{\sqrt[n]{a_1 a_2 \cdots a_n}} = 1$. By Lemma 3.1, the above holds. The equality holds iff $\frac{a_1}{\sqrt[n]{a_1 a_2 \cdots a_n}} = \frac{a_2}{\sqrt[n]{a_1 a_2 \cdots a_n}} = \cdots = \frac{a_n}{\sqrt[n]{a_1 a_2 \cdots a_n}}$, i.e., $a_1 = a_2 = \cdots = a_n$.

3. $A(n) \leq Q(n)$. Let $c = \frac{a_1 + a_2 + \cdots + a_n}{n}$ and $a_i = c + \alpha_i, \forall i = 1$. Then

$$\begin{aligned} a_1 + a_2 + \cdots + a_n &= nc + \alpha_1 + \alpha_2 + \cdots + \alpha_n \\ &= a_1 + a_2 + \cdots + a_n + (\alpha_1 + \alpha_2 + \cdots + \alpha_n) \end{aligned}$$

   So, $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 0$.

$$\begin{aligned} \sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}} &= \sqrt{\frac{(c + \alpha_1)^2 + (c + \alpha_2)^2 + \cdots + (c + \alpha_n)^2}{n}} \\ &= \sqrt{\frac{nc^2 + 2c(\alpha_1 + \alpha_2 + \cdots + \alpha_n) + \alpha_1^2 + \alpha_2^2 + \cdots + \alpha_n^2}{n}} \\ &= \sqrt{\frac{nc^2 + \alpha_2^2 + \cdots + \alpha_n^2}{n}} \\ &= \sqrt{c^2 + \frac{\alpha_2^2 + \cdots + \alpha_n^2}{n}} \\ &\geq \sqrt{c^2} = c \end{aligned}$$

where the third equality follows from $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 0$ and $c = \frac{a_1+a_2+\cdots+a_n}{n}$.

Lemma 3.1 is not involved with the proof of $A(n) \leq Q(n)$. □

Note. AM≥GM≥HM can be proved using induction; see Theorem 1.2.2 of Jixiu Chen et al. Mathematical Analysis, third edition.

3.1 An inequality about the difference between AM and GM

Given $a, b, c \in \mathbf{R}_+$, show $3(\frac{a+b+c}{3} - \sqrt[3]{abc}) \geq 2(\frac{a+b}{2} - \sqrt{ab})$.

Proof. After simple algebra, it suffices to show

$$c + 2\sqrt{ab} \geq 3\sqrt[3]{abc}$$

If we consider $2\sqrt{ab}$ as $\sqrt{ab} + \sqrt{ab}$ on the LHS, we can use AM-GM to get

$$c + \sqrt{ab} + \sqrt{ab} \geq 3\sqrt[3]{\sqrt{ab} \cdot \sqrt{ab} \cdot c} = 3\sqrt[3]{abc}.$$

This completes our proof. □

This result can be easily shown for the general case, namely

$$(n+1)(\frac{a_1^2 + a_2^2 + \cdots + a_{n+1}^2}{n+1} - \sqrt[n+1]{a_1 a_2 \cdots a_{n+1}}) \geq n(\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n} - \sqrt[n]{a_1 a_2 \cdots a_n}).$$

3.2 Power mean inequality

Refer to Page 106 of Xicheng Peng et al. Exploring Inequalities. 2016. This will be done later.

3.3 Weighted power mean inequality

Refer to Page 108 of Xicheng Peng et al. Exploring Inequalities. 2016. This will be done later.

3.4 Applications

Given positive reals a, b, c and $a + b + c = 1$, show that $ab + bc + ca \leq \frac{1}{3}$.

Proof. Since the geometric mean is not less than the average mean, we have

$$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a+b+c}{3} = \frac{1}{3} \Longleftrightarrow a^2 + b^2 + c^2 \geq \frac{1}{3}.$$

Also we have

$$ab + bc + ca = \frac{(a+b+c)^2 - (a^2 + b^2 + c^2)}{2} = \frac{1 - (a^2 + b^2 + c^2)}{2} \leq \frac{1 - 1/3}{2} = \frac{1}{3},$$

as desired. □